

Is LACK OF STANDARDS DELAYING ADOPTION Of IoT STANDARDS And HAMPERING SECURITY

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This article is broadly based on a panel discussion on security threats in the IoT paradigm, conducted during IEW 2018

Just as tech entrepreneurs jumped onto the app-development band wagon a couple of years ago, the Internet of Things (IoT) device development became the new startup mantra. Yet, with all the frenetic activity in this domain, has the cart overtaken the horse? Of what use are thousands of innovative devices if these are not secure? In fact, IoT adoption itself is severely hampered by people's concerns about security.

The words that keep cropping up in every discussion on the IoT are heterogeneous and exponential. A smart city best illustrates the heterogeneity of the IoT. It requires service providers from diverse industries offering electricity supply, sanitation, waste management, healthcare, transportation and the all-pervasive Wi-Fi to interact with each other. Typically, these industries work in isolation, but now these need smart and secure interfaces to communicate with each other.

At India Electronics Week (IEW) 2018, during a panel discussion on IoT security, Binoy C.S., director of digital transformation practice (ICT) at Frost and Sullivan (F&S), gave the audience a sense of the exponential growth in IoT devices. F&S has been tracking the IoT for over eight years. Currently, the firm believes that, there are 12.5 billion connected devices globally. By 2020, the figure is expected to go up to 20 billion and by 2024, to 80 billion—adding up to 10 devices per human.

At this point, it is important for us to remember that a smart city is very much a work in progress. And its eventual realisation hinges on a number of factors—the most important being security.

In 2017, Forbes reported in a survey it conducted in the

US, related to the IoT, that 98 per cent of respondents believed that security breaches would go up as devices proliferated. But monitoring devices and interfaces is only possible with globally-accepted standards. At the moment, those standards do not exist.

Just how daunting this task of standardisation is, was pointed out by Sunil David, regional director, AT&T, in the following comparison. After the PC market matured, it was driven by about five major manufacturers.

So is the case with telecom. A few big players like Vodafone, AT&T and Airtel operate across the globe. Therefore arriving at standardisation involves working with fewer players. Forming consortiums for standards and security is relatively simple. But in the IoT universe, we already have over 10,000 companies jostling for space. And these are all operating in diverse industries.

Where does one begin with standards

So who should take the lead? Is it governments or firms like Intel, ARM and AMD, whose components go into the billions of electronic products churned out every day, along with telecom giants like AT&T, Vodafone and Airtel?

This is an important consideration because there are two aspects of security, often in conflict with each other. On one hand is personal privacy and security. These concerns were brought into focus when several private parties moved Indian courts because of their worries about how safe citizen data is, with Aadhaar.

Then, there is the issue of national security. In 2017, Wikileaks leaked documents purportedly demonstrating how the CIA (US' Central Intelligence Agency) hacks smartphones, computer operating systems, messaging apps and Internet-linked televisions to spy on people—its own citizens as well as those from other countries. So how

